MTH 101 TRICKS, SHORTCUTS AND CALCULATOR METHODS

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NOTE: WE ARE NOT GIVING ALL THESE CALCULATOR TIPS AND TRICKS TO MAKE YOU LAZY BUT TO SAVE YOU SOME TIME IN YOUR TESTS AND EXAMINATIONS. THANK YOU

TOPICS TREATED:

1. Vector (physics)
2. Operation on real number/Polynomials
3. Sequence and series (AP&GP)
4. Partial fraction
5. Theory of quadratics
6. Matrices
7. Set

1. TO RESET THE CALCUTOR TO DEFAULT MODE

The calculation mode can be returned to the initial default by pressing the [**SHIFT**] [**CLR**] [**2**] [**MODE**] [=] [**AC**] or simply [**MODE**] [**1**] (**COMP**).
All data currently in the calculator memory is cleared when this key sequence is used: [**SHIFT**] **[CLR**] [**3**][=] [**AC**] .

2. TO STORE A VALUE ON THE CALCULATOR

To store number or answer on the calculator memory, the following procedure must be followed

➢ Punch the number to be stored and press **[=];**

➢ Press **[SHIFT] [STO]** followed by the alphabet you want to store the number.

The stored number could be recalled by pressing **[RCL]** followed by the alphabet that the number was stored.
EX: To store the number **23.6** unto the calculator memory, the following key sequence must be followed:

➢ Enter **23.6 [=] [SHIFT] [STO] [A].** The number has been stored as **A.**

To retrieve the stored number, the following key sequence must be followed:

➢ Press **[RCL] [A].**

3. TO PICK A MODE FUNCTION

Mode Description	Key Operation	Notation
Basic Arithmetic	[MODE] [1]	COMP
Complex Number	[MODE] [2]	CMPLX
Statistical Calculation	[MODE] [3]	REG
Base-N Calculation	[MODE] [4]	BASE
Equation	[MODE] [5]	EQN
Matrix Calculation	[MODE] [6]	MAT
Table Calculation	[MODE] [7]	TABLE
Vector Calculation	[MODE] [8]	VCT
Decimal Place	[SHIFT] [MODE] [6]	Fix
Significant Figures	[SHIFT] [MODE] [7]	Sci

VECTOR (PHYSICS)

Examples:

1. If A and B are vectors specified by A= (2, -3, 5) and B=(5,3,-4), find A+3B (a)(17,5,-7) (b) (6,5,4) (c) 17,6,-7(d) none
2. Find the absolute value of vector -3i+2j+5k
   a) 4 (b) -37 (c) √38 (d) none
3. Given A=(3,4,5) and B=(6,8,10). Find A.B
   (a) 100 (b) 86 (c) 35 (d) 59
4. Find the magnitude of │A│=(3,4,5)… (a) 7.07 (b) 4.65 (c) 3.87
5. Find the angle between A= (3i+2j+6k) and B= (4i-3j+k).. (a) 60° (b) 30° (c) 90° (d) 0°

Solutions

Q1:

i. Put your calculator in vector mode. To do that, (Mode+8)
ii. Store your data (press Vct A)
iii. Since Vct A is 3-dimensional vector, press 1
iv. Put in your data using = sign to go to next data i.e 2=, -3= 5=
v. Your data is stored automatically
vi. Press AC
vii. Press shift+5, pick 2 (data), pick 2 (Vct B) to fill for second vector
viii. Since Vct B is also 3-dimensional vector, press 1
ix. Put in your data using = sign to go to next data i.e 5=, 3= -4=
x. Your data is saved. Press AC
xi. Question is A+3B. Press shift+5, pick 3 (Vct A), press+, Press 3, Press shift+5 again, press 4 (Vct B)…… VctA+3VctB……press =
xii. Ans is 17…………… Do you get that?????? Good…

Q2: (i.e magnitude)

i. Reset the calc …..shift, 9, 3, AC
ii. Put your calculator in vector mode. To do that, (Mode+8)
iii. Store your data (press Vct A)
iv. Since Vct A is 3-dimensional vector, press 1
v. Put in your data using = sign to go to next data i.e -3=, 2= 5=
vi. Your data is stored automatically

vii. Press AC

viii. Press shift, hyp button(Abs), Shift again, 5, pick 3 (Vct A) to fill and close) …… Abs(VctA)

ix. Press =

x. Ans is 6.16 or √38 (press SD to change to decimal) …………… Do you get that?????? Good…

<u>Q3:</u>

i. Put your calculator in vector mode. To do that, (Mode+8)

ii. Store your data (press Vct A)

iii. Since Vct A is 3-dimensional vector, press 1

iv. Put in your data using = sign to go to next data i.e 3=, 4= 5=

v. Your data is stored automatically

vi. Press AC

vii. Press shift+5, pick 2 (data), pick 2 (Vct B) to fill for second vector

viii. Since Vct B is also 3-dimensional vector, press 1

ix. Put in your data using = sign to go to next data i.e 6=, 8= 10=

x. Your data is saved. Press AC

xi. Question is A.B Press shift+5, pick 3 (Vct A), press shift, 5 again, Press 7(Dot), Press shift+5 again, press 4 (Vct B)…… (VctA.3VctB)……press =

xii. Ans is 100…………… Do you get that?????? Good…

<u>Q4:</u> (i.e magnitude)

i. Reset the calc …..shift, 9, 3, AC

ii. Put your calculator in vector mode. To do that, (Mode+8)

iii. Store your data (press Vct A)

iv. Since Vct A is 3-dimensional vector, press 1

v. Put in your data using = sign to go to next data i.e 3=, 4= 5=

vi. Your data is stored automatically

vii. Press AC

viii. Press shift, hyp button(Abs), Shift again, 5, pick 3 (Vct A) to fill and close) …… Abs(VctA)

ix. Press =

x. Ans is 7.07 …………… Do you get that?????? Good…

<u>Q5:</u>

I. Reset the calc …..shift, 9, 3, AC

II. Put your calculator in vector mode. To do that, (Mode+8)
III. Store your data (press Vct A)
IV. Since Vct A is 3-dimensional vector, press 1
V. Put in your data using = sign to go to next data i.e 3=, 2= -6=
VI. Your data is stored automatically
VII. Press AC
VIII. Press shift+5, pick 2 (data), pick 2 (Vct B) to fill for second vector
IX. Since Vct B is also 3-dimensional vector, press 1
X. Put in your data using = sign to go to next data i.e 4=, -3= 1=
XI. Your data is saved. Press AC

Normally, the workings should be:

$$\text{Cos } \phi = \frac{A.B}{|AB|} = \frac{0}{35.7}$$

$$\phi = \cos^{-1} \frac{0}{35.7}$$

XII. Find A.B Press shift+5, pick 3 (Vct A), press shift, 5 again, Press 7(Dot), Press shift+5 again, press 4 (Vct B)…… (VctA.3VctB)……press =
XIII. Ans is 0…………… Do you get that?????? Good…
XIV. Find A.B Press shift+5, pick 3 (Vct A), press shift, 5 again, Press 7(Dot), Press shift+5 again, press 4 (Vct B)…… (VctA.3VctB)……press =
XV. Find magnitude of AB, Press shift, hyp button(Abs), Shift again, 5, pick 3 (Vct A) to fill Shift again, 5, pick 4 (Vct B)…… Abs(VctAxVctB)….. press =
XVI. Ans is 35.7
XVII. Press AC
XVIII. Reset the calc …..shift, 9, 3, AC
XIX. Press Shift, Cos to get $\cos^{-1}$, put fraction symbol i.e $\cos^{-1} (\frac{0}{35.7})$…… press =
Ans ... 90
Always reset your calculator when you are done…

Operation on real number/Polynomials

Examples

1. Given two polynomials f(x) = $8x^2 - 3x^2$ and q(x) = $5x^4 - x^2 + 1$ what is f(x)×q(x) (a) $40x^7 - 15x^6 - 8x^5 + 3x^4 + 8x^3 - 3x^2$(b) $40x^8 - 15x^7 - 8x^6 + 3x^5 + 8x^4 - 3x^3$(c) $42x^7 - 18x^6 - 12x^5 + 8x^4 + 8x^3 - 6x^2$(d) $30x^7 - 15x^6 - 8x^5 + 3x^4 + 6x^3 - 3x^2$

LONG DIVISION AND MULTIPLICATION OF POLYNOMIALS

1. The remainder upon division of the polynomial $8x^3 - 2x^2 - 3x + 2$ by 4x-3 is? A. $\frac{13}{4}$ B.$\frac{17}{4}$ C. 2 D. none
2. The polynomial $3x^2 - 7x - 6$ divides $x - 3$ to give which of these (a) $3x + 3$ (b)$3x + 4$ (c)$3x + 2$ (d) $2x + 3$
3. Which of these is a factor of the polynomial $2x^2 - 7x - 4$ **(a)** $x - 4$ (b) $2x - 2$ (c)$3x - 1$ (d) $x - 3$
4. $x^2 - 4 \times 2x - 1$ will give us which of these polynomials (a) $2x^3 + x^2 + 8x + 4$ **(b)** $2x^3 - x^2 - 8x + 4$ (c) $2x^3 + x^2 - 8x + 4$ (d) $4x^3 - x^2 - 8x + 2$

ADDITION AND SUBTRACTION OF POLYNOMIALS

1. Given two polynomials Q(x) = $7x^5 - 4x^3 + 2x^2 - 8x + 3$ and P(x) = $4x^5 - 9x^3 + 11x^2 + 15x - 7$, Q(x) – P(x) = **(a)** $3x^5 + 5x^3 - 9x^2 - 23x + 10$ (b) $3x^4 + 5x^3 - 9x^2 - 23x + 10$ (c) $6x^5 + 5x^3 - 9x^2 - 20x + 10$ (d) $11x^5 - 13x^3 + 13x^2 + 7x - 4$
2. Given two polynomials Q(x) = $7x^5 - 4x^3 + 2x^2 - 8x + 3$ and P(x) = $4x^5 - 9x^3 + 11x^2 + 15x - 7$, P(x) + Q(x) = (a) $3x^5 + 5x^3 - 9x^2 - 23x + 10$ (b) $3x^4 + 5x^3 - 9x^2 - 23x + 10$ (c) $6x^5 + 5x^3 - 9x^2 - 20x + 10$ **(d)** $11x^5 - 13x^3 + 13x^2 + 7x - 4$

CALCULATOR METHOD FOR SOLVING LOND DIVISION OF POLYNOMIALS

Question 1

Steps

Make x the subject of the formula first

I. Clear the calculator by pressing Shift then 9 then 3 then = twice
II. Store the value of x in the calculator as ¾ by pressing the fraction button on the calculator ($\frac{\square}{\square}$) then put 3 as numerator and 4 as denominator then press shift followed by the **RCL** button and the the alphabet **x** it will then show that ¾ has been stored as x on the calculator then press **AC**
III. Enter the polynomial by pressing 8 followed by **ALPHA** then **x** then – 2 then **ALPHA** **x** then – 3 then **ALPHA** **x** then + 2 and then +
IV. The answer will come out as 2 which is the correct answer option (c)

SEQUNCES AND SERIES (AP and GP)

A.P

1. The 6th term of an arithmetic progression is 12 and the 30th term is 180 find A. the first term, B. the 52nd term, C. if the nth term is 250 find the nth term D. what is the common difference of the sequence

Steps

I. Put the calculator in statistic mode(MODE then 3 Stat)
II. Pick option 2 which represents A.P
III. Input the values, X represents the number of terms Y represents the number

x	y
6	12
30	180

IV. Press AC
V. Press shift and 1, press option 5 REG, press option 5 ($\hat{y}$)
VI. To find 1st term press 1($\hat{y}$) and the answer will be –23
VII. To find 52nd term press 52($\hat{y}$) and the answer will be 334

VIII. To find the nth term press Shift then 1 then 5, then go for 4 (ẍ) Press 250 (ẍ) the answer will be 40. To confirm if it is correct press 40 (ŷ). It should give you 250.

IX. To find the Common Difference, press the term after 6^{th} term which is 7 that is 7(ŷ) -12 and the answer will be 7.

SUM OF A.P

Example: The 6^{th} term of an A.P is 12, and the 30^{th} term is 180, calculate the sum of the first 60 terms

Steps

I. Reset your calculator by pressing Shift, 9 and 3 and AC
II. Put the calculator in statistic mode(MODE then 3 Stat)
III. Pick option 2 which represents A.P
IV. Input the values, X represents the number of terms Y represents the number

x	y
6	12
30	180

V. Press AC
VI. Press shift and log $, to\ bring \sum press\ alpha + x$ press Shift 1 then option 5 (ŷ) then press Shift X to get comma, i.e. $\sum(x\,\hat{y}\,, 1, 60)$ then press =. The answer should be 11010

G.P

Example: The 4^{th} term of a Geometric progression is 256, and the 8^{th} term is 16, find A. The 3^{rd} term B. the 1^{st} term C. common ratio D. nth term if the nth term is 32.

Steps

I. Reset your calculator by pressing Shift, 9 and 3 and AC
II. Put the calculator in statistic mode(MODE then 3 Stat)
III. Pick option 6 which represents G.P
IV. Input the values, X represents the number of terms Y represents the number

x	y
4	256
8	16

V. Press AC
VI. Press shift and 1, the option 5 and then option 5 again (ŷ)

VII. To find 3rd term press 3(ŷ) and the answer will be 512

VIII. To find 1st term press 1(ŷ) and the answer will be 2048

IX. To find the nth term press Shift then 1 then 5, then go for 4 (ẍ) Press 32 (ẍ) the answer will be 7. To confirm if it is correct press 7 (ŷ). It should give you 32.

X. To find the Common Ratio, press the term after 4th term which is 5 that is 5(ŷ) / 256 and the answer will be 0.5

SUM OF G.P

Example: The 1st term of a G.P is 7, and the 5th term is 567, calculate the sum of the first 9 terms

<u>Steps</u>

I. Reset your calculator by pressing Shift, 9 and 3 and AC

II. Put the calculator in statistic mode(MODE then 3 Stat)

III. Pick option 6 which represents G.P

IV. Input the values, X represents the number of terms Y represents the number

x	y
1	7
5	567

V. Press AC

VI. Press shift and log $, to\ bring \sum press\ alpha + x,$ press Shift 1 then option 5 (ŷ) then press Shift X to get comma, i.e. $\sum(x\, ŷ\, , 1{,}9)$ then press =. The answer should be 68887

PARTIAL FRACTIONS

Example: resolve into partial fractions $\frac{3x+5}{2x^2-5x-3}$ A. $\frac{2}{x+3} - \frac{1}{2x-1}$ B. $\frac{1}{x+3} - \frac{2}{2x-1}$ C. $\frac{2}{x-3} - \frac{1}{2x-1}$ D. none

Steps

I. Use option testing method. Put x = 1 and test your options till you get the one that is the same with the question.

THEORY OF QUADRATICS

The key sequence for quadratic equation is given as [**MODE**] [**5**][**3**] . The option ‘3’ is used for solving quadratic equations and option ‘4’ is for Cubic Equations thus the option 3 and 4 are the polynomial modes.
HINTS; choose option 3 if you have a quadratic equation (X^2) and option 4 if you have a cubic equation (X^3).
A. Cubic equations $\boldsymbol{ax^3+bx^2+cx+d=0}$ are also solved the same way using the calculator.
B. The general equation of quadratic equation is given as $ax^2+bx+c=0$.

Example 1

Find the roots of the equation $2x^2-6x+4=0$

Procedure

➢ Press **[MODE] [5] [3]**

➢ Enter the coefficients in the equation ($=2$,$b=-6$ and $c=4$)

➢ **[2] [=] [−6] [=] [4] [=]**

Result:$_1=2$ [=] $x_2=1$

Example 2

Solve $3x^3+2x^2+4x+5=0$

Procedure

➢ Press **[MODE] [5] [4]**

➢ Enter the coefficients in the equation ($=3$,$b=2$, $c=4$ and $d=5$)

➢ [3] [=] [2] [=] [4] [=] [5] [=]

Result:$_1=-1$ [=] $x_2=16+1.2802i$ [=] $x_3=16-1.2802i$

TRY IT OUT!

1. Solve the equation $x^3+x^2-81x-81=0$
Result: $x_1=9$ [=] $x_2=-1$ [=] $x_3=-9$

2. Find the roots of the quadratic equation $2x^3+5x^2-14x-8=0$
Result: $x_1=-3$ [=] $x_2=3$ [=] $x_3=1$

3. Solve the equation $x^2+3x-10=0$
Result: $x_1=2$ [=] $x_2=-5$

SIMULTANEOUS EQUATION (EQUATIONS IN TWO/THREE VARIABLES)

To turn on the simultaneous equation function: **[MODE] [5].** Option **1** and **2** are the simultaneous equation modes. Select option 1 if the equation has two unknown variables or option 2 if it has three unknown variables. Here we are interested in two unknown variables. The procedure is exactly the same for three unknown variables.

1: [anX+bnY=cn] (Simultaneous Equation In Two Variables)

2: [anX+bnY+CnZ=dn] (Simultaneous Equation In Three Variables)

Example 1

Find the value of x and y in the following equations:

$\mathbf{2x}+\mathbf{y}=\mathbf{5}$

$\mathbf{2x}+\mathbf{2y}=\mathbf{6}$

a1: Coefficient of x in equation 1=2

b1: Coefficient of y in equation 1=1

c1: The constant in equation 1=5

a2: Coefficient of x in equation 2=2

b2: Coefficient of y in equation 2=2

c2: The constant in equation 2=6

Result: ($\mathbf{x}$ =**2** [=] **y**=**1**) **(Bravo!).**

CHECK!

Putting $\mathbf{x}=\mathbf{2}$ and $\mathbf{y}=\mathbf{1}$ into the equations:

$\mathbf{2x}+\mathbf{y}=\mathbf{5}$ and $\mathbf{2x}+\mathbf{2y}=\mathbf{6}$

2(2)+1=5 and 2(2)+2(1)=6 **(Interesting!)**

Example 2:

Solve the set of equations

$x+2y+z=4$……… (1)

$3x-4y-2z=2$…… (2)

$5x+3y+5z=-1$… (3)

Procedure

- [MODE] [5] [2]
- Row 1: [**ENTER**] **1** [=] **2** [=] **1** [=] **4** [=]
- Row 2: [**ENTER**] **3** [=] −**4** [=] −**2** [=] **2** [=]
- Row 3: [**ENTER**] **5** [=] **3** [=] **5** [=] −**1** [=]

Result:(x=2 [=] y=3 [=] z=−4)

TRY THESE

1. Find the value of x, and z if $\mathbf{5x}-\mathbf{2y}+\mathbf{3z}=\mathbf{16}$; $\mathbf{2x}+\mathbf{3y}-\mathbf{5z}=\mathbf{2}$ and $\mathbf{4x}-\mathbf{5y}+\mathbf{6z}=\mathbf{7}$

Result x=3,=7 and z=5

2. Find the value of a and b $\mathbf{a}+\mathbf{b}=-\mathbf{4}$ and $\mathbf{4a}+\mathbf{b}=\mathbf{2}$

Result a=2 and b=−6

ALPHA AND BETA PART

Example 1: If $\alpha\ and\ \beta$ be the roots of the quadratic equation $3x^2 - x - 1 = 0$ find the new quadratic equation whose roots are $\alpha^2\ and\ \beta^2$ A. $9x^2 - 7x + 1 = 0$ B. $2x^2 - 3x - 5 = 0$ C. $x^2 - 4x - 3 = 0$ D. $2x^2 + 3x - 1 = 0$

Steps

I. Press MODE then select option 5 i.e. EQN
II. Select 3 for quadratic EQN

III. Enter your values .i.e. A = 3, B = -1, C = -1 and press =
IV. After getting your X1 and X2 convert to decimal by press S-D
V. Store X1 which is α *and* $X2$ *which is* β
VI. Press shift then 9 then 3 then AC to clear the calculator
VII. Press 0.767 which is α or X1, to store it press shift, RCL button
VIII. Press A to store it as A
IX. Press -0.435 which is β or X2, and store it as B
X. To bring out the saved α *and* β press Alpha + A and square i.e. A^2 and do for B^2 too
XI. Check the options that will give you A^2 and B^2
XII. Select 3 for quadratic EQN
XIII. Enter your values foe each
XIV. Check for each value of your X1 or X2 of the options that resembles A^2 and B^2 of your question

MATRICES

Addition, Subtraction, Multiplication, Determinant, inverse, transpose

Example 1: add $\begin{matrix} 2 & 3 \\ 4 & 2 \\ 1 & 0 \end{matrix}$ and $\begin{matrix} 1 & 6 \\ 2 & 4 \\ 3 & 5 \end{matrix}$

Steps

I. Clear the calculator
II. Press mode then option 6 then 1, for MatA and select the order (3x2) option 2
III. Enter your values
IV. Press AC
V. Press shift + 4 pick the option Data (2) to enter values for MatB, same order
VI. Enter your values
VII. Press AC
VIII. To find the addition or subtraction press Shift + 4 then press 3 then press + and then press shift + 4 again and 4 for MatB. i.e. MatA+MatB and then = to get your answer.

Example 2: find the determinant $|A|$ if A is a 2x2 matrix $\begin{matrix} 4 & 2 \\ 6 & 5 \end{matrix}$

Steps

I. Clear the calculator
II. Press mode then option 6 then 1, for MatA and select the order (2x2) option 5

III. Enter your values
IV. Press AC
V. Press shift + 4 pick the option Det (7)
VI. Press shift + 4 to select MatA which is option 3. i.e. det(MatA)
VII. Press =

Example 3: Find the inverse and transpose of A $\begin{matrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{matrix}$

Steps

I. Clear the calculator
II. Press mode then option 6 then 1, for MatA and select the order (3x3) option 1
III. Enter your values
IV. Press AC
V. Press shift + 4 pick the option Trn (8) for transpose
VI. Press shift + 4 to select MatA which is option 3. i.e. Trn(MatA)
VII. Press =

For the inverse press

I. Press shift + 4 to select MatA which is option 3, then press $\mathbf{x^{-1}}$ button
II. Press =

SET THEORY

GENERAL TIPS AND TRICKS

1. You must **start with the inner circle before going to other parts of the Venn diagrams**

2. Two sets **A and B are said to be equal** only if both the sets have the same and exact number of elements

3. Empty or null set **has no elements**

4. **The intersection denoted as A n B is the set of elements common** to both A and B

5. The **union of sets A and B, written as A u B Is the set of elements that appear in either A OR B**

6. The difference of sets A and B, is the set of elements **belonging to set A** and **NOT to set B**

FOR TWO VENN DIAGRAM
1. For how many like **EITHER** of the subject question (i.e how many student studies both), TRICKS TO BE USED, **n(AUB)=n(A)+n(B) – n(AnB)**

Example: In a class of 100 students, 35 like science, 45 like maths, 10 like both. How many like both? Try it with the trick

2. **For how many** like **NEITHER** of the subject question (i.e how many student doesn't study or like any of the subject), TRICKS TO BE USED, **Total students n(AuB) minus both subject n(AnB)**
Example: In a class of 100 students, 35 like science, 45 like maths, 10 like both. How many like neither? Try it with the trick

3. **Only one subject only**….. Tricks to be used
n(A) = **Total students n(AuB) - n(B) - both subject n(AnB)…………………. Same way with n(B)**

FOR THREE VENN DIAGRAM
Ex: A survey of 100 students asked about their languages they speak , 50 speaks Yoruba, 35 speaks Igbo, 25 speaks Hausa, 20 speaks Yoruba and Igbo and 15 speaks Igbo and Hausa and 12 speaks Hausa and Yoruba

NOTE: The moment they say ‘’**at least’’, NONE IS ALWAYS ZERO**. SO NO NEED TO DISTURB YOURSELF

1. **Yoruba only**; n(Yoruba) – (all + n (Y n I)+ n(Y n H))
2. **Hausa only**: n(Hausa) – (all+ n(H n Y)+ n(H n I))
3. **Igbo only**: n(Igbo) – (all+ n(I n Y)+ n(I n H))
4. **Yoruba and Hausa only**: n(Y n H) – All
5. **Yoruba and Igbo only**: n(Y n I) - All
6. **Hausa and Igbo**: n(H n I) - All
7. **NONE** (I.e how many students study none of the subjects or read none of the magazines as the case may be)

None (ABC) = A-B -C + Both AB + Both BC + Both CA - All (ABC) + Total (universal)

8. **ALL** (I.e how many students study all subjects or read all magazines as the case may be)
All (ABC) = A-B -C + Both AB + Both BC + Both CA + Total (universal) - None (ABC)

9. TOTAL (universal) = A+B+C – Both AB – Both BC – Both CA + None (ABC) + All (ABC)

10. Exactly one: n(A) + n(B)+ n(C) – 2n(AnB) - 2n(AnC) -2n(BnC) + 3(AnBnC)

11. Exactly two: n(AnB) + n(BnC) + n(AnC) - 3(AnBnC)

12. Exactly three: n(AnBnC)

What is the Set Theory?
Set Theory is the process of collection of objects, sets which are known as elements or
numbers.

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Let’s understand set A and B, the two sets.

● The number of elements which are prevalent either in set A or B has is represented by
$n(A\cup B)$

● $n(A\cap B)$ is the number of elements which are present in sets ***A*** and set ***B***.

$n(A\cup B) = n(A) + (n(B) – n(A\cap B)$

If we talk about Set A, B & C, the formula is:

$n(A\cup B\cup C) = n(A) + n(B) + n(C) – n(A\cap B) – n(B\cap C) – n(C\cap A) + n(A\cap B\cap C)$

Set Theory Formulas
Stated below are the important set theory formulas-

Set Theory Formulas on Properties

● Commutativity:

○ $A \cap B = B \cap A$

○ $A \cup B = B \cup A$
● Associativity:
○ $A \cap (B \cap C) = (A \cap B) \cap C$
○ $A \cup (B \cup C) = (A \cup B) \cup C$
● Distributivity:
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
● Idempotent Law:
○ $A \cap A = A$
○ $A \cup A = A$
● Law of $\varnothing$ and $\cup$:
○ $A \cap \varnothing = \varnothing$
○ $U \cap A = A$
○ $A \cup \varnothing = A$
○ $U \cup A = U$

Sets Theory Formulas of Difference of Sets

● $A - A = \varnothing$
● $B - A = B \cap A'$
● $B - A = B - (A \cap B)$
● $n(A \cup B) = n(A - B) + n(B - A) + n(A \cap B)$
● $n(A - B) = n(A \cup B) - n(B)$
● $n(A - B) = n(A) - n(A \cap B)$
● $(A - B) = A$ if $A \cap B = \varnothing$
● $(A - B) \cap C = (A \cap C) - (B \cap C)$
● $A \Delta B = (A-B) \cup (B- A)$
● $n(A') = n(\cup) - n(A)$

Sets Theory Formulas of Complement Sets

● Law of Double complementation: $(A')' = A$
● Laws of Empty set and Universal Set: $\varnothing' = \cup$ and $\cup' = \varnothing$
● Complement Law : $A \cup A' = U$, $A \cap A' = \varnothing$ and $A' = U - A$
● De Morgan's Laws: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$

Contact these for any enquiry

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